

AWS Relational Database Service(RDS)

1.Create MariaDB DB on EC2.

Objective: Create a MariaDB database server on an Amazon EC2 instance, configure it securely, create a database and user, and verify that the database is working correctly.

Prerequisites

Before starting, ensure you have:

- AWS Account
- EC2 Instance (Amazon Linux 2023)
- Key Pair (.pem file)
- Security Group
- Internet Gateway
- Public IP Address
- SSH Client (Git Bash/PowerShell/Terminal)

Step 1: Launch EC2 Instance

Login to AWS Console

Go to

EC2

Click

Launch Instance

Configure

Option	Value
Name	MariaDB-Server
AMI	Amazon Linux 2023
Instance Type	t2.micro

Option	Value
Key Pair	Existing/New
Network	Default or Custom VPC
Public IP	Enable
Security Group	Create New

Step 3: Connect to EC2

Step 4: Update the Server

```
sudo dnf update -y
```

Step 5: Check Available MariaDB Packages

```
sudo dnf list mariadb*
```

Step 6: Install MariaDB Server

```
sudo dnf install mariadb105-server -y
```

Verify installation

```
mariadb --version
```

Example output

```
mariadb Ver 15.1 Distrib 10.5.xx
```

Step 7: Start MariaDB Service

```
sudo systemctl start mariadb
```

Step 8: Enable MariaDB at Boot

```
sudo systemctl enable mariadb
```

Step 9: Verify Service Status

`sudo systemctl status mariadb`

Expected output

Active: active (running)

```
Complete!
[root@ip-172-31-21-46 ~]# mariadb --version
mariadb Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper
[root@ip-172-31-21-46 ~]# sudo systemctl start mariadb
[root@ip-172-31-21-46 ~]# sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@ip-172-31-21-46 ~]# sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-08-01 12:54:40 UTC; 19s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 27491 (mariadb)
   Status: "Taking your SQL requests now..."
     Tasks: 13 (limit: 1059)
    Memory: 66.2M
       CPU: 467ms
   CGroup: /system.slice/mariadb.service
           └─27491 /usr/libexec/mariadb --basedir=/usr

Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: The>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: you>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: Aft>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: ab)>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: See>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: Ple>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: The>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: Con>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal mariadb-prepare-db-dir[27447]: htt>
Aug 01 12:54:40 ip-172-31-21-46.ec2.internal systemd[1]: Started mariadb.servic>
```

Step 10: Secure MariaDB Installation

Run: `sudo mysql_secure_installation`

Typical responses:

Enter current password: Press Enter

Switch to unix_socket authentication? Y

Change root password? Y

Enter new password: *****

Remove anonymous users? Y

Disallow root login remotely? Y

Remove test database? Y

Reload privilege tables? Y

```

root@ip-172-31-21-46:~
[root@ip-172-31-21-46 ~]# sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] Y
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] Y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] Y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] Y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] Y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] Y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
[root@ip-172-31-21-46 ~]#

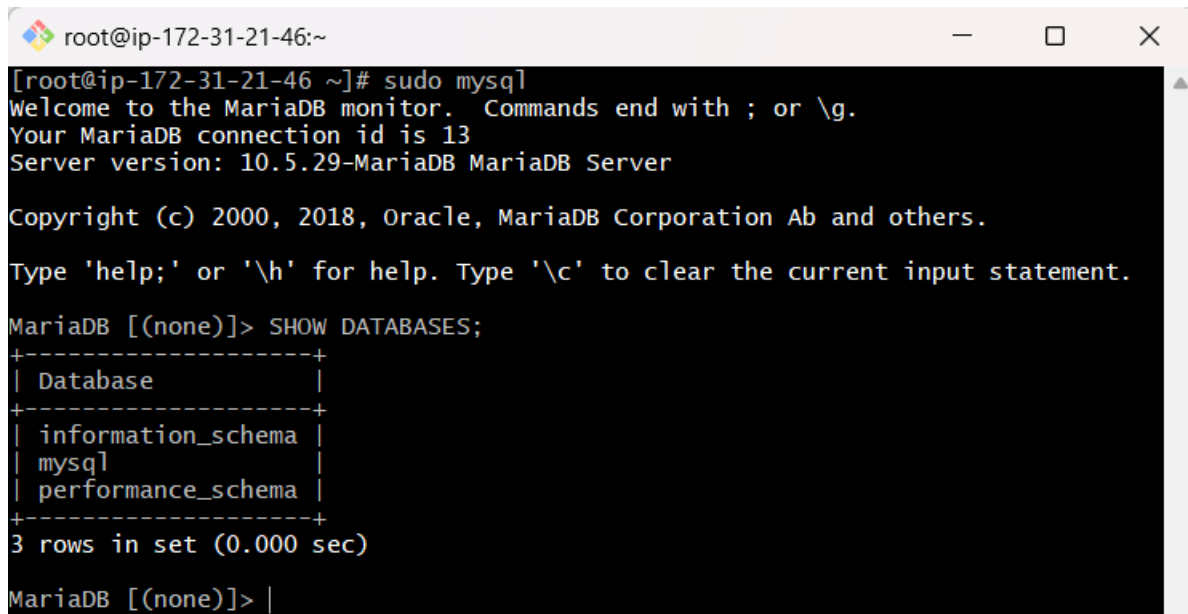
```

Step 11: Login to MariaDB

sudo mysql or mysql -u root -p

Step 12: Show Existing Databases

SHOW DATABASES;



```
root@ip-172-31-21-46:~  
[root@ip-172-31-21-46 ~]# sudo mysql  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 13  
Server version: 10.5.29-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| information_schema |  
| mysql |  
| performance_schema |  
+-----+  
3 rows in set (0.000 sec)  
MariaDB [(none)]> |
```

2. Insert some dummy data.

Step 1: Create a Database

CREATE DATABASE employee_db;

Verify: SHOW DATABASES;

Step 3: Select the Database

USE employee_db;

Step 4: Create the employees Table

```
CREATE TABLE employees (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    department VARCHAR(100),  
    designation VARCHAR(100),  
    salary DECIMAL(10,2),  
    joining_date DATE  
);
```

Verify the table: SHOW TABLES;

```
MariaDB [(none)]> CREATE DATABASE employee_db;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| employee_db |
| information_schema |
| mysql |
| performance_schema |
+-----+
4 rows in set (0.000 sec)

MariaDB [(none)]> USE employee_db;
Database changed
MariaDB [employee_db]> CREATE TABLE employees (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> name VARCHAR(100) NOT NULL,
  -> department VARCHAR(100),
  -> designation VARCHAR(100),
  -> salary DECIMAL(10,2),
  -> joining_date DATE
  -> );
Query OK, 0 rows affected (0.008 sec)

MariaDB [employee_db]> show tables;
-> show tables;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MariaDB server version for the right syntax to use near ':
show tables' at line 1
MariaDB [employee_db]> show tables;
+-----+
| Tables_in_employee_db |
+-----+
| employees |
+-----+
1 row in set (0.000 sec)

MariaDB [employee_db]> |
```

Step 5: Insert Dummy Data

Step 6: View the Data

SELECT * FROM employees;

```
MariaDB [employee_db]> INSERT INTO employees (name, department, designation, salary, joining_date) VALUES
  -> ('Raj Kumar', 'DevOps', 'DevOps Engineer', 65000.00, '2023-01-15'),
  -> ('Priya Sharma', 'HR', 'HR Executive', 48000.00, '2022-11-10'),
  -> ('Ravi Kumar', 'Development', 'Software Engineer', 72000.00, '2021-08-25')
),
  -> ('Anjali Reddy', 'QA', 'QA Engineer', 55000.00, '2024-02-12'),
  -> ('Suresh Babu', 'Finance', 'Accountant', 60000.00, '2020-09-05');
Query OK, 5 rows affected (0.003 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [employee_db]> SELECT * FROM employees;
+-----+-----+-----+-----+-----+-----+
| id | name      | department | designation      | salary  | joining_date |
+-----+-----+-----+-----+-----+-----+
| 1 | Raj Kumar | DevOps     | DevOps Engineer  | 65000.00 | 2023-01-15 |
| 2 | Priya Sharma | HR        | HR Executive     | 48000.00 | 2022-11-10 |
| 3 | Ravi Kumar | Development | Software Engineer | 72000.00 | 2021-08-25 |
| 4 | Anjali Reddy | QA        | QA Engineer      | 55000.00 | 2024-02-12 |
| 5 | Suresh Babu | Finance    | Accountant       | 60000.00 | 2020-09-05 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [employee_db]> |
```

Step 7: Verify the Table Structure

```
MariaDB [employee_db]> DESCRIBE employees;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
department	varchar(100)	YES		NULL	
designation	varchar(100)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
joining_date	date	YES		NULL	

```
6 rows in set (0.001 sec)

MariaDB [employee_db]> |
```

Step 8: Count the Records

```
MariaDB [employee_db]> SELECT COUNT(*) FROM employees;
```

COUNT(*)
5

```
1 row in set (0.000 sec)

MariaDB [employee_db]> |
```

3. Take the backup of dummy data on EC2.

Objective: Create a backup of the MariaDB database and store it on the EC2 instance.

Step 1: Verify the Database

Login to MariaDB: # sudo mysql

Show databases: > SHOW DATABASES;

Select the database: > USE employee_db;

Verify the data: > SELECT * FROM employees;

Exit MariaDB: > EXIT;

```
MariaDB [employee_db]> DESCRIBE employees;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
department	varchar(100)	YES		NULL	
designation	varchar(100)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
joining_date	date	YES		NULL	

```
6 rows in set (0.001 sec)

MariaDB [employee_db]> SELECT COUNT(*) FROM employees;
```

COUNT(*)
5

```
1 row in set (0.000 sec)

MariaDB [employee_db]> EXIT;
Bye
[root@ip-172-31-21-46 ~]#
```

Step 2: Create a Backup Directory

Create a directory to store database backups.

```
# mkdir -p ~/db-backups
```

```
[root@ip-172-31-21-46 ~]# mkdir -p ~/db-backups
[root@ip-172-31-21-46 ~]# ls
db-backups
[root@ip-172-31-21-46 ~]#
```

Step 3: Take the Database Backup

If you're using **root with socket authentication** (common on Amazon Linux):

```
root@ip-172-31-21-46:~
[root@ip-172-31-21-46 ~]# ls
db-backups
[root@ip-172-31-21-46 ~]# sudo mariadb-dump employee_db > ~/db-backups/employee_db_backup.sql
[root@ip-172-31-21-46 ~]# ls
db-backups
[root@ip-172-31-21-46 ~]#
```

4. Launch MariaDB RDS instance.

Objective: Create an Amazon RDS MariaDB database instance, connect to it, create a database and table, insert sample data, and verify connectivity.

Prerequisites

- AWS Account
- VPC (default or custom)

- At least one public subnet (or private subnet with a bastion host)
- Security Group
- EC2 instance (optional, for testing connectivity)
- MySQL/MariaDB client installed

Step 1: Open the RDS Console

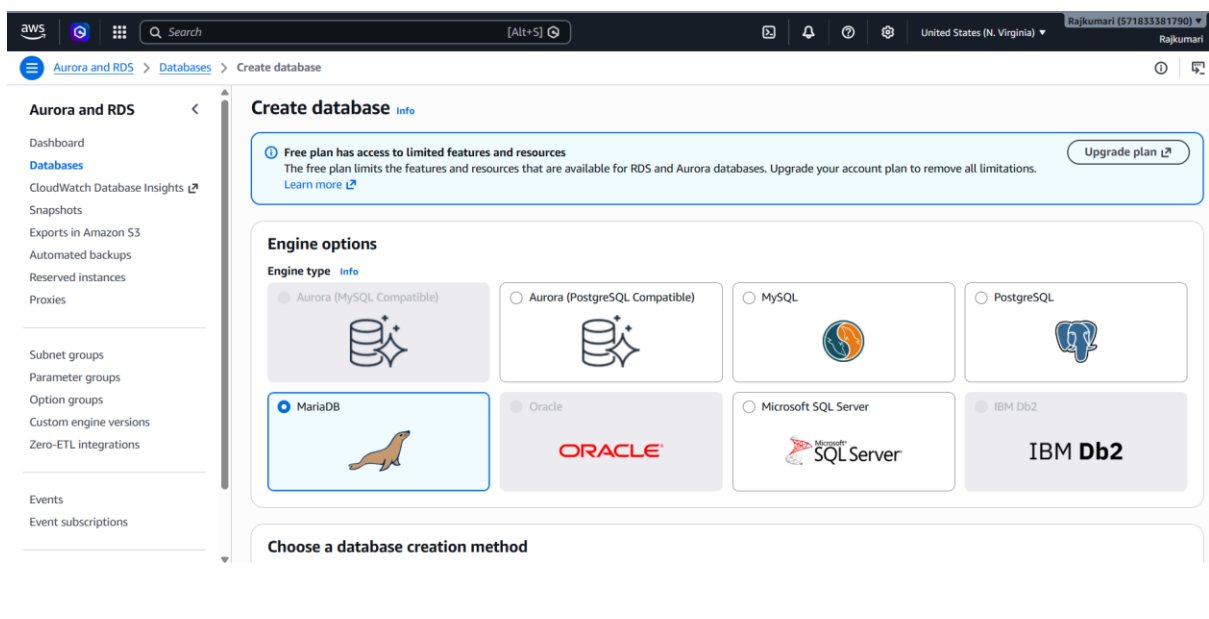
1. Log in to the AWS Management Console.
2. Search for **RDS**.
3. Open the **Amazon RDS** service.
4. Click **Create database**.

Step 2: Choose Database Creation Method

- Select **Standard Create**.

Step 3: Select Database Engine

- Engine type: **MariaDB**
- Edition: **MariaDB Community**
- Engine version: Select the latest available version (or the version specified by your assignment).



Step 4: Configure the DB Instance

Setting	Value
DB Instance Identifier	mariadb-rds
Master Username	admin
Master Password	Password@123 (example; use a strong password)
Confirm Password	Same as above

Step 5: Instance Configuration

Option	Value
DB Instance Class	db.t3.micro (or db.t4g.micro if eligible in your region)
Storage Type	General Purpose SSD (gp3)
Allocated Storage	20 GB
Storage Autoscaling	Optional

Step 6: Connectivity

Configure the network:

Setting	Value
VPC	Default or your custom VPC
DB Subnet Group	Default
Public Access	Yes (for lab/demo)
VPC Security Group	Create new or select existing

Note: For production, databases are usually **not publicly accessible**. They are placed in private subnets and accessed through application servers or bastion hosts.

Step 7: Security Group

Allow inbound traffic:

Type	Port	Source
MySQL/Aurora	3306	Your IP address (or the EC2 security group that will connect)

Step 8: Additional Configuration

Setting	Value
Initial Database Name	employee_db
Backup Retention	7 days (default)
Monitoring	Default
Performance Insights	Optional
Deletion Protection	Disable for lab environments

Step 9: Create the Database

Click **Create database**.

The status will progress through:

Creating

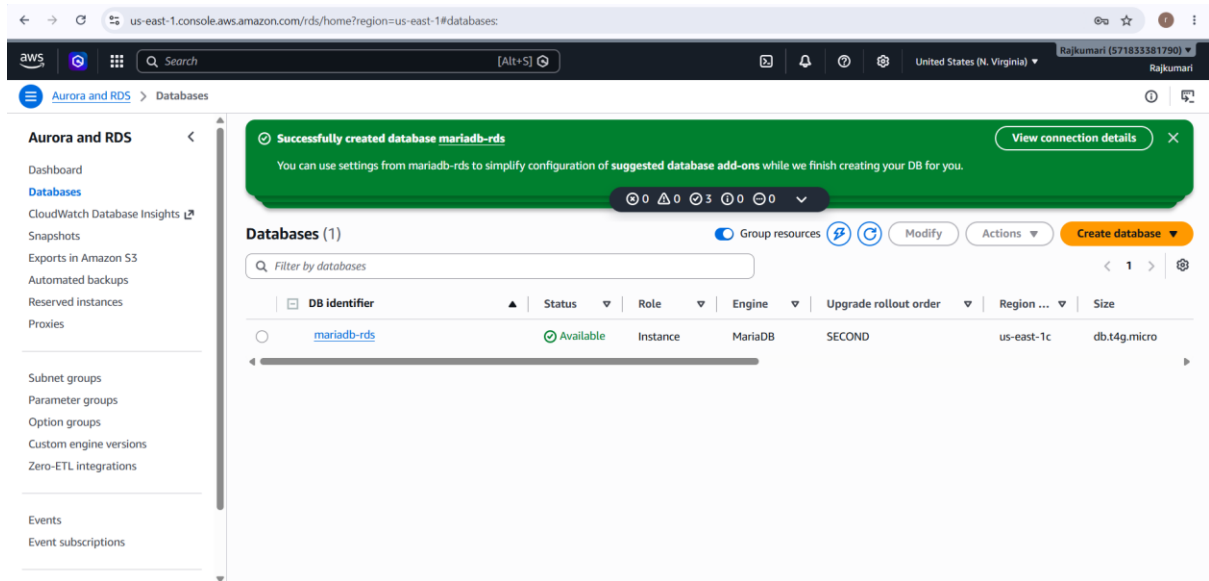
↓

Backing-up

↓

Available

This may take **5–15 minutes**.



Step 10: Copy the Endpoint

When the status becomes **Available**:

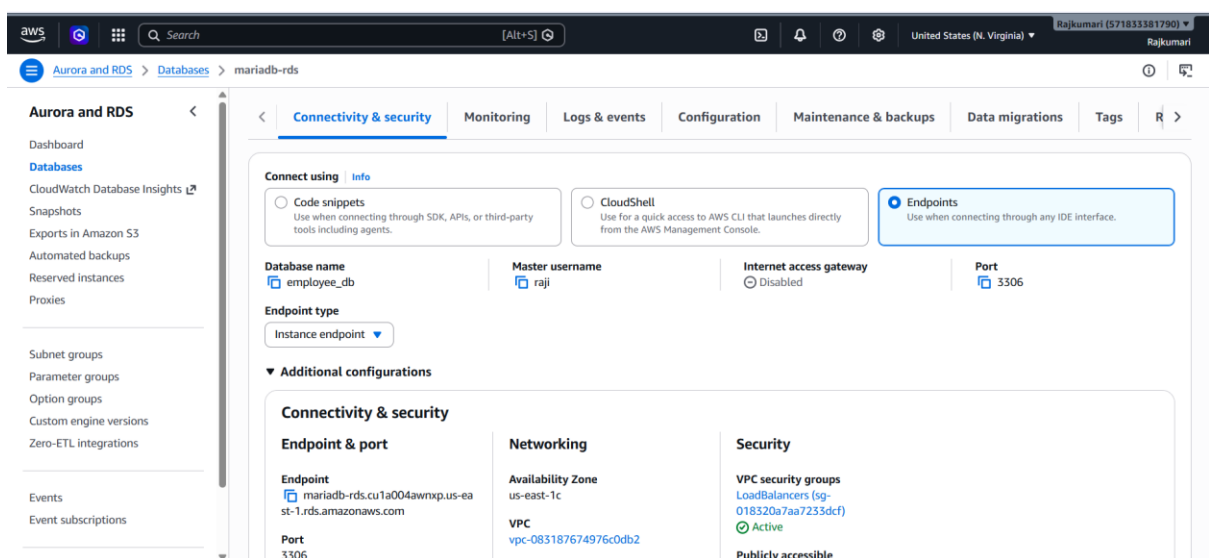
1. Click the DB instance.
2. Under **Connectivity & security**, copy the **Endpoint**.

Example:

mariadb-rds.abcdefghijkl.us-east-1.rds.amazonaws.com

Port:

3306



```

[root@ip-172-31-21-46 ~]# mysql -h mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com -u raji -p
Enter password:
ERROR 3159 (08004): Connections using insecure transport are prohibited while --require_secure_transport=ON.
[root@ip-172-31-21-46 ~]# wget https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
--2026-08-01 14:31:21-- https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
Resolving truststore.pki.rds.amazonaws.com (truststore.pki.rds.amazonaws.com)... 13.226.209.35, 13.226.209.117, 13.226.209.110, ...
Connecting to truststore.pki.rds.amazonaws.com (truststore.pki.rds.amazonaws.com)|13.226.209.35|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 165408 (162K) [binary/octet-stream]
Saving to: 'global-bundle.pem'

global-bundle.pem      100%[=====>] 161.53K  --.-KB/s   in 0.002s

2026-08-01 14:31:21 (63.6 MB/s) - 'global-bundle.pem' saved [165408/165408]

[root@ip-172-31-21-46 ~]# ls -l global-bundle.pem
-rw-r--r--. 1 root root 165408 Jul 24 2025 global-bundle.pem
[root@ip-172-31-21-46 ~]# mysql -h mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com \
-u raji \
-p \
--ssl-ca=global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 52
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW STATUS LIKE 'Ssl_cipher';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| Ssl_cipher    | TLS_AES_256_GCM_SHA384 |
+-----+-----+
1 row in set (0.003 sec)

MariaDB [(none)]> \C
exit
exit
exit
Bye
[root@ip-172-31-21-46 ~]# mysql -h mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com -u raji -p
Enter password:
ERROR 3159 (08004): Connections using insecure transport are prohibited while --require_secure_transport=ON.
[root@ip-172-31-21-46 ~]# mysql -h mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com \
-u raji \
-p \
--ssl-ca=global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 62
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

```

```

root@ip-172-31-21-46:~
--ssl-ca=global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 62
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
MariaDB [(none)]>
MariaDB [(none)]>
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| employee_db |
| information_schema |
| innodb |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.007 sec)

MariaDB [(none)]> USE employee_db;
Database changed
MariaDB [employee_db]> CREATE TABLE employees (
->   id INT AUTO INCREMENT PRIMARY KEY,
->   name VARCHAR(100),
->   department VARCHAR(100),
->   salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.028 sec)

MariaDB [employee_db]> INSERT INTO employees (name, department, salary)
-> VALUES
-> ('Raj Kumar','DevOps',65000.00),
-> ('Priya Sharma','HR',48000.00),
-> ('Ravi Kumar','Development',72000.00),
-> ('Anjali Reddy','QA',55000.00),
-> ('Suresh Babu','Finance',60000.00);
Query OK, 5 rows affected (0.009 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [employee_db]> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name      | department | salary |
+----+-----+-----+-----+
| 1  | Raj Kumar | DevOps     | 65000.00 |
| 2  | Priya Sharma | HR        | 48000.00 |
| 3  | Ravi Kumar | Development | 72000.00 |
| 4  | Anjali Reddy | QA        | 55000.00 |
| 5  | Suresh Babu | Finance    | 60000.00 |
+----+-----+-----+-----+
5 rows in set (0.001 sec)

```

```

Complete!
[root@ip-172-31-21-46 ~]# telnet mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com 3306
Trying 172.31.19.234...
Connected to mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com.
Escape character is '^['.
V
11.8.8-MariaDB-logMp>#4y+b1kIk_{k'~hnmYmysql_native_passwordConnection closed by foreign host.
[root@ip-172-31-21-46 ~]#

```

5. Migrate database from EC2 to RDS.

EC2 MariaDB → mariadb-dump → RDS MariaDB → Restore

Since you already created employee_db on RDS, I will show the proper migration process from scratch.

Step 1: Verify Database on EC2

Login to your EC2 MariaDB:

```
mysql -u root -p
```

Check databases:

SHOW DATABASES;

Example:

employee_db

Check tables:

USE employee_db;

SHOW TABLES;

```
root@ip-172-31-21-46:~  
[root@ip-172-31-21-46 ~]# mysql -u raji -p  
Enter password:  
ERROR 1045 (28000): Access denied for user 'raji'@'localhost' (using password: YES)  
[root@ip-172-31-21-46 ~]# mysql -u raji -p  
Enter password:  
ERROR 1045 (28000): Access denied for user 'raji'@'localhost' (using password: YES)  
[root@ip-172-31-21-46 ~]# mysql -u root -p  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 29  
Server version: 10.5.29-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> SELECT User, Host FROM mysql.user;  
+-----+-----+  
| User      | Host      |  
+-----+-----+  
| mariadb.sys | localhost |  
| mysql      | localhost |  
| root       | localhost |  
+-----+-----+  
3 rows in set (0.001 sec)  
  
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database      |  
+-----+  
| employee_db   |  
| information_schema |  
| mysql         |  
| performance_schema |  
+-----+  
4 rows in set (0.000 sec)  
  
MariaDB [(none)]> use employee_db;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
MariaDB [employee_db]> show tables;  
+-----+  
| Tables_in_employee_db |  
+-----+  
| employees              |  
+-----+  
1 row in set (0.000 sec)  
  
MariaDB [employee_db]> |
```

Step 2: Take Backup from EC2 MariaDB

Exit MariaDB:

exit;

Run: mariadb-dump employee_db > employee_db_backup.sql

Step 3: Connect to RDS MariaDB

Because your RDS requires SSL:

```
root@ip-172-31-21-46:~  
[root@ip-172-31-21-46 ~]# mariadb-dump employee_db > employee_db_backup.sql  
[root@ip-172-31-21-46 ~]# ls -lh employee_db_backup.sql  
-rw-r--r--. 1 root root 2.5K Aug  1 15:54 employee_db_backup.sql  
[root@ip-172-31-21-46 ~]# mysql \  
-h mariadb-rds.cu1a004awnxp.us-east-1.rds.amazonaws.com \  
-u raji \  
-p \  
--ssl-ca=global-bundle.pem  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 206  
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> |
```

Step 5: Create Database in RDS

Inside RDS MariaDB:

```
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 206  
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> CREATE DATABASE employee_migration;  
Query OK, 1 row affected (0.007 sec)  
  
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| employee_db |  
| employee_migration |  
| information_schema |  
| innodb |  
| mysql |  
| performance_schema |  
| sys |  
+-----+  
7 rows in set (0.001 sec)  
  
MariaDB [(none)]> exit  
Bye  
[root@ip-172-31-21-46 ~]# |
```

Step 6: Restore Database into RDS

Run from EC2 Linux:

Enter RDS password.

Step 7: Verify Migration

Connect again:

Check: SHOW DATABASES;

Use database: USE employee_migration;

Check tables: SHOW TABLES;

Check data : SELECT * FROM employees;

```
root@ip-172-31-21-46:~  
[root@ip-172-31-21-46 ~]# mysql \-h mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com \-u raji \-p \--ssl-ca=global-bundle.pem \employee_migration < employee_db_backup.sql  
Enter password:  
[root@ip-172-31-21-46 ~]# mysql \-h mariadb-rds.cula004awnxp.us-east-1.rds.amazonaws.com \-u raji \-p \--ssl-ca=global-bundle.pem  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 220  
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| employee_db |  
| employee_migration |  
| information_schema |  
| innodb |  
| mysql |  
| performance_schema |  
| sys |  
+-----+  
7 rows in set (0.001 sec)
```

```
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| employee_db |  
| employee_migration |  
| information_schema |  
| innodb |  
| mysql |  
| performance_schema |  
| sys |  
+-----+  
7 rows in set (0.001 sec)  
  
MariaDB [(none)]> USE employee_migration;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
MariaDB [employee_migration]> SHOW TABLES;  
+-----+  
| Tables_in_employee_migration |  
+-----+  
| employees |  
+-----+  
1 row in set (0.001 sec)  
  
MariaDB [employee_migration]> SELECT * FROM employees;  
+-----+-----+-----+-----+-----+-----+  
| id | name | department | designation | salary | joining_date |  
+-----+-----+-----+-----+-----+-----+  
| 1 | Raj Kumar | DevOps | DevOps Engineer | 65000.00 | 2023-01-15 |  
| 2 | Priya Sharma | HR | HR Executive | 48000.00 | 2022-11-10 |  
| 3 | Ravi Kumar | Development | Software Engineer | 72000.00 | 2021-08-25 |  
| 4 | Anjali Reddy | QA | QA Engineer | 55000.00 | 2024-02-12 |  
| 5 | Suresh Babu | Finance | Accountant | 60000.00 | 2020-09-05 |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.001 sec)  
  
MariaDB [employee_migration]> |
```

6. Install MySQL DB on EC2.

1. Connect to EC2 Instance

2. Update Packages

3. Install MySQL Server

For Amazon Linux 2026:

Step 1: Install MySQL Community Repository

Download MySQL repository package: `sudo dnf install`

`https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm -y`

Step 2: Import MySQL GPG Key: `sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023`

Step 3: Check MySQL Repository: `dnf repolist | grep mysql`

```
root@ip-172-31-21-46:~
[root@ip-172-31-21-46 ~]# sudo dnf install https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm -y
Last metadata expiration check: 5:04:55 ago on Sat Aug 1 12:53:33 2026.
mysql80-community-release-el9-1.noarch.rpm                               60 kB/s | 10 kB      00:00
Dependencies resolved.
=====
Package                        Architecture  Version      Repository      Size
=====
Installing:
mysql80-community-release      noarch        el9-1        @commandline    10 k
=====
Transaction Summary
=====
Install 1 Package

Total size: 10 k
Installed size: 5.7 k
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Installing                : mysql80-community-release-el9-1.noarch 1/1
  Verifying                 : mysql80-community-release-el9-1.noarch 1/1
Installed:
mysql80-community-release-el9-1.noarch

Complete!
[root@ip-172-31-21-46 ~]# sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023
[root@ip-172-31-21-46 ~]# dnf repolist | grep mysql
mysql-connectors-community      MySQL Connectors Community
mysql-tools-community          MySQL Tools Community
mysql80-community              MySQL 8.0 Community Server
[root@ip-172-31-21-46 ~]# |
```

Step 5: Install MySQL Server

Now install: `sudo dnf install mysql-community-server -y`

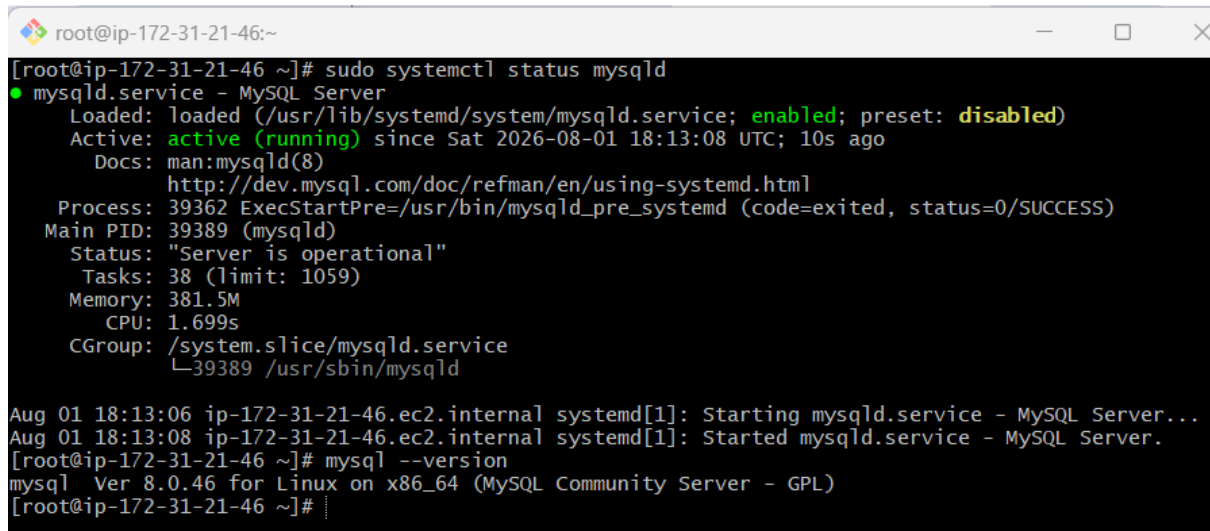
Step 6: Start MySQL Service

Enable at boot: `sudo systemctl enable mysqld`

Start: `sudo systemctl start mysqld`

Check: `sudo systemctl status mysqld`

Step 7: Check MySQL Version: `mysql --version`

A terminal window titled 'root@ip-172-31-21-46:~' with standard window controls. The terminal shows the command 'sudo systemctl status mysqld' being executed. The output indicates that the 'mysqld.service - MySQL Server' is loaded and active (running) since August 1, 2026, at 18:13:08 UTC. It provides details about the process (PID 39362), main PID (39389), status ('Server is operational'), tasks (38), memory (381.5M), CPU usage (1.699s), and CGroup. Below this, it shows log messages from systemd[1] stating 'Starting mysqld.service - MySQL Server...' and 'Started mysqld.service - MySQL Server.'. Finally, the command 'mysql --version' is executed, showing 'mysql Ver 8.0.46 for Linux on x86_64 (MySQL Community Server - GPL)'.

```
[root@ip-172-31-21-46 ~]# sudo systemctl status mysqld
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-08-01 18:13:08 UTC; 10s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
   Process: 39362 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
  Main PID: 39389 (mysqld)
    Status: "Server is operational"
     Tasks: 38 (limit: 1059)
    Memory: 381.5M
       CPU: 1.699s
   CGroup: /system.slice/mysqld.service
           └─39389 /usr/sbin/mysqld

Aug 01 18:13:06 ip-172-31-21-46.ec2.internal systemd[1]: Starting mysqld.service - MySQL Server...
Aug 01 18:13:08 ip-172-31-21-46.ec2.internal systemd[1]: Started mysqld.service - MySQL Server.
[root@ip-172-31-21-46 ~]# mysql --version
mysql Ver 8.0.46 for Linux on x86_64 (MySQL Community Server - GPL)
[root@ip-172-31-21-46 ~]#
```

7. Launch MySQL RDS image.

Step 1: Open Amazon RDS

1. Sign in to the AWS Management Console.
2. Search for **RDS**.
3. Open the **RDS** service.
4. Click **Create database**.

Step 2: Choose Database Creation Method

- Select **Standard Create**.

Step 3: Choose Engine

- **Engine type:** MySQL
- **Edition:** MySQL Community

Step 4: Choose Engine Version

Select the latest stable version available (for example, **MySQL 8.4.x LTS** or the version required by your assignment).

Step 5: Templates

For a lab or assignment:

- **Free tier** (if available), or
 - **Dev/Test**
-

Step 6: Settings

- **DB instance identifier:** mysql-rds
 - **Master username:** admin (or as required)
 - **Master password:** Create a strong password.
 - Confirm the password.
-

Step 7: Instance Configuration

- **DB instance class:** db.t3.micro (or Free Tier eligible if available)
-

Step 8: Storage

- **Storage type:** General Purpose SSD (gp3)
 - **Allocated storage:** 20 GB (or the minimum allowed)
-

Step 9: Connectivity

- **VPC:** Select your VPC.
- **Public access:** **Yes** (for learning/testing) or **No** if you will connect only from EC2.
- **VPC Security Group:** Select an existing security group or create a new one.

Security Group Inbound Rule

Allow MySQL access:

Type	Port	Source
MySQL/Aurora	3306	Your EC2 Security Group (recommended) or your EC2 private IP

Step 10: Database Authentication

- Password authentication
-

Step 11: Additional Configuration

- **Initial database name:** employee_db (optional)
 - **Backup retention:** 1days (or default)
 - Leave the remaining settings as default unless your assignment specifies otherwise.
-

Step 12: Create Database

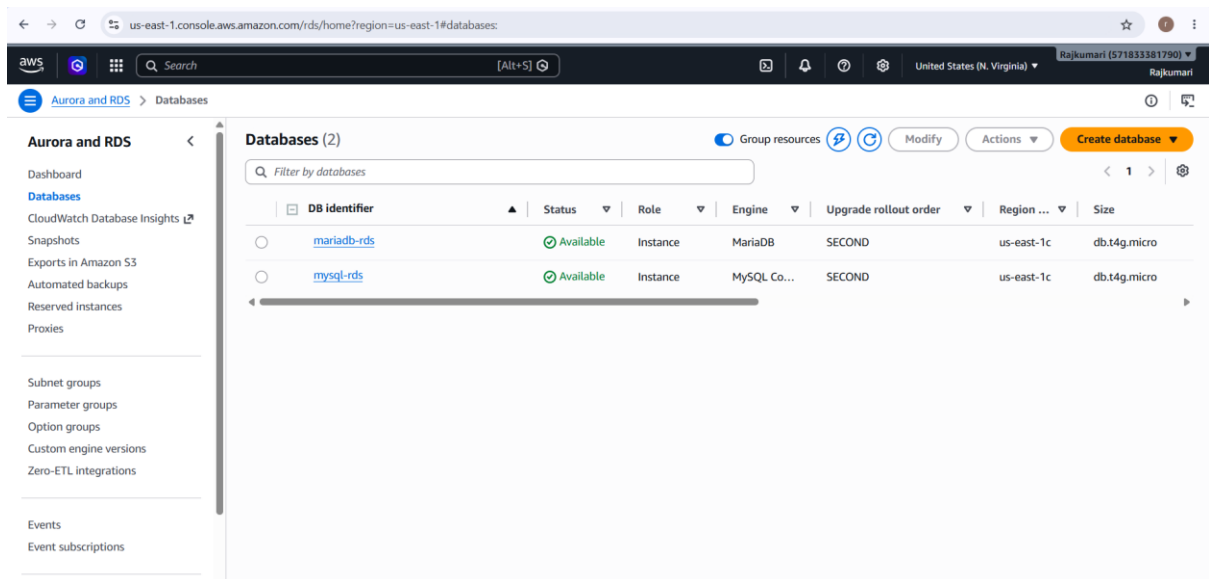
Click **Create database**.

Wait for the status to change from:

Creating

to

Available



Step 13: Copy the Endpoint

After the instance becomes **Available**:

1. Open the DB instance.
2. Copy the **Endpoint**.

Step 14: Connect from EC2

Install the MySQL client on your EC2 if needed, then connect:

```

root@ip-172-31-20-27:~
mysql-community-common      x86_64  8.0.46-1.el9  mysql80-community  558 k
mysql-community-libs       x86_64  8.0.46-1.el9  mysql80-community  1.5 M

Transaction Summary
-----
Install 4 Packages

Total download size: 6.9 M
Installed size: 96 M
Downloading Packages:
(1/4): mysql-community-common-8.0.46-1.el9.x86_  29 MB/s | 558 kB    00:00
(2/4): mysql-community-client-plugins-8.0.46-1.  41 MB/s | 1.4 MB    00:00
(3/4): mysql-community-libs-8.0.46-1.el9.x86_64  53 MB/s | 1.5 MB    00:00
(4/4): mysql-community-client-8.0.46-1.el9.x86_  53 MB/s | 3.5 MB    00:00
-----
Total                                           102 MB/s | 6.9 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                                     : 1/1
  Installing : mysql-community-client-plugins-8.0.46-1.el9.x86_64 1/4
  Installing : mysql-community-common-8.0.46-1.el9.x86_64         2/4
  Installing : mysql-community-libs-8.0.46-1.el9.x86_64          3/4
  Running scriptlet: mysql-community-libs-8.0.46-1.el9.x86_64    3/4
  Installing : mysql-community-client-8.0.46-1.el9.x86_64        4/4
  Running scriptlet: mysql-community-client-8.0.46-1.el9.x86_64  4/4
  Verifying   : mysql-community-client-8.0.46-1.el9.x86_64        1/4
  Verifying   : mysql-community-client-plugins-8.0.46-1.el9.x86_64 2/4
  Verifying   : mysql-community-common-8.0.46-1.el9.x86_64        3/4
  Verifying   : mysql-community-libs-8.0.46-1.el9.x86_64          4/4

Installed:
mysql-community-client-8.0.46-1.el9.x86_64
mysql-community-client-plugins-8.0.46-1.el9.x86_64
mysql-community-common-8.0.46-1.el9.x86_64
mysql-community-libs-8.0.46-1.el9.x86_64

Complete!
[root@ip-172-31-20-27 ~]# mysql --version
mysql Ver 8.0.46 for Linux on x86_64 (MySQL Community Server - GPL)
[root@ip-172-31-20-27 ~]# mysql -h mysql-rds.cula004awnxp.us-east-1.rds.amazonaws.com -u raji -p
Enter password:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 43
Server version: 8.4.9 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |

```

Step 15: Verify

Inside MySQL:

SHOW DATABASES;

```

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.04 sec)

mysql> |

```

8. Configure Multi - AZ.

Objective: Configure Multi-AZ for an Amazon RDS MySQL instance to improve high availability and automatic failover. With Multi - AZ enabled, Amazon RDS keeps a standby database in a different Availability Zone (AZ). If the primary database or its AZ becomes unavailable, RDS automatically switches to the standby instance.

What is Multi - AZ?

Multi-AZ is a high availability feature of Amazon RDS.

- One **Primary** database instance handles all application traffic.
- One **Standby** database instance is created in another Availability Zone.
- The standby instance is **not used for read traffic**.
- Data is replicated **synchronously** from the primary to the standby.
- If the primary instance fails, Amazon RDS performs an **automatic failover** to the standby.

Configure Multi-AZ During Database Creation

Step 1: Open Amazon RDS

1. Log in to the AWS Management Console.
2. Go to **RDS**.
3. Click **Databases**.
4. Select your database:

mysql-rds

Step 2: Modify the Database

Click the **Modify** button in the top-right corner.

Step 3: Availability & Durability

Scroll down to the **Availability & durability** section.

Look for **Multi-AZ deployment**.

If your instance supports it, select:

- ☒ **Create a standby instance (Multi-AZ DB instance)**

or simply enable **Multi-AZ deployment**, depending on the console version.

Step 4: Continue

Scroll to the bottom and click **Continue**.

Step 5: Apply the Change

You'll be asked when to apply the modification.

Choose one of the following:

- **Apply immediately** (recommended for your assignment)
- Apply during the next maintenance window

Select:

Apply immediately

Then click:

Modify DB Instance

- ➔ **You selected the Free Tier template** when creating the RDS instance. Multi-AZ is not available with the Free Tier configuration.
- ➔ **Your DB instance class does not support Multi-AZ** (or your account is using a configuration where Multi-AZ isn't offered).
- ➔ Your AWS account may have service limitations in the current Region.

9. Take backup of DB and restore the DB.

Objective: To create a backup of an Amazon RDS MySQL database and restore it to a new RDS database instance.

Step 1: Connect to the RDS MySQL Database

From your EC2 instance:

Step 2: Create a database: `CREATE DATABASE company_db;`

Verify: `SHOW DATABASES;`

Step 3: Use the database: `USE company_db;`

Step 4: Create a table: `CREATE TABLE employees (`

`id INT AUTO_INCREMENT PRIMARY KEY,`

`name VARCHAR(100),`

`department VARCHAR(50),`

`salary DECIMAL(10,2)`

);

Step 4: Insert data: INSERT INTO employees (name, department, salary)

VALUES

('Raj Kumar','DevOps',65000),

('Priya','HR',50000),

('Ravi','Development',70000);

Step 5: Verify: SELECT * FROM employees;

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.04 sec)

mysql> CREATE DATABASE company_db;
Query OK, 1 row affected (0.05 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| company_db |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)

mysql> USE company_db;
Database changed
mysql> CREATE TABLE employees (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   name VARCHAR(100),
  ->   department VARCHAR(50),
  ->   salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO employees (name, department, salary)
  -> VALUES
  -> ('Raj Kumar','DevOps',65000),
  -> ('Priya','HR',50000),
  -> ('Ravi','Development',70000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name   | department | salary |
+----+-----+-----+-----+
| 1  | Raj Kumar | DevOps    | 65000.00 |
| 2  | Priya    | HR        | 50000.00 |
| 3  | Ravi     | Development | 70000.00 |
+----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

Step 6: Take the backup

Exit MySQL: exit;

Run: mysqldump -h <RDS Endpooint> -u raji -p company_db > company_db_backup.sql

Step 7: Restore the backup

Create a new database: mysql -h mysql-rds.cu1a004awnxp.us-east-1.rds.amazonaws.com -u raji -p

10

```
Database changed
mysql> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name   | department | salary |
+----+-----+-----+-----+
| 1  | Raj Kumar | DevOps      | 65000.00 |
| 2  | Priya    | HR          | 50000.00 |
| 3  | Ravi     | Development | 70000.00 |
+----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM company_db_restore;
ERROR 1146 (42S02): Table 'company_db_restore.company_db_restore' doesn't exist
mysql> SELECT * FROM company_db_restore;
ERROR 1146 (42S02): Table 'company_db_restore.company_db_restore' doesn't exist
mysql> show tables;
+-----+
| Tables_in_company_db_restore |
+-----+
| employees                     |
+-----+
1 row in set (0.02 sec)

mysql> SELECT * FROM company_db_restore.employees;
+----+-----+-----+-----+
| id | name   | department | salary |
+----+-----+-----+-----+
| 1  | Raj Kumar | DevOps      | 65000.00 |
| 2  | Priya    | HR          | 50000.00 |
| 3  | Ravi     | Development | 70000.00 |
+----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

10. Create read replica.

Objective: Create a **Read Replica** for an existing Amazon RDS MySQL database to improve **read performance** by offloading read queries from the primary database.

What is a Read Replica?

A **Read Replica** is a read-only copy of your primary RDS database.

- It receives data from the primary database using **asynchronous replication**.
- Applications can send **SELECT** queries to the replica.
- All **INSERT**, **UPDATE**, and **DELETE** operations must be performed on the primary database.
- It is mainly used for **read scaling**, **reporting**, and **analytics**.

Prerequisites

Before creating a Read Replica:

- ☒ Primary RDS MySQL instance is in **Available** state.
- ☒ Automated backups are enabled (backup retention period must be greater than 0).
- ☒ You have permission to create RDS instances.

Step-by-Step Process

Step 1: Open Amazon RDS

1. Sign in to the AWS Management Console.
2. Open **Amazon RDS**.
3. Click **Databases**.

Step 2: Select the Primary Database

Select your database:

mysql-rds

Step 3: Create Read Replica

Click:

Actions

Select:

Create Read Replica

Step 4: Configure the Replica

Fill in the following:

Setting	Example
DB Instance Identifier	mysql-rds-replica
DB Instance Class	db.t3.micro (or another supported class)
Availability Zone	No Preference

Setting	Example
Storage Type	gp3
Public Access	Same as primary

Step 5: Create the Replica

Click:

Create Read Replica

AWS begins creating the replica.

Status:

Creating

After several minutes:

Available

Create read replica

You are creating a replica DB instance from a source DB instance. This new DB instance will have the source DB instance's DB security groups and DB parameter groups.

Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#) [Upgrade plan](#)

Settings

Replica source
Source DB instance identifier
mysql-rds
Role: Instance

DB instance identifier
This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (for example, mydbinstance).
mysql-rds-replica

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

Hide filters

☐ Include previous generation classes

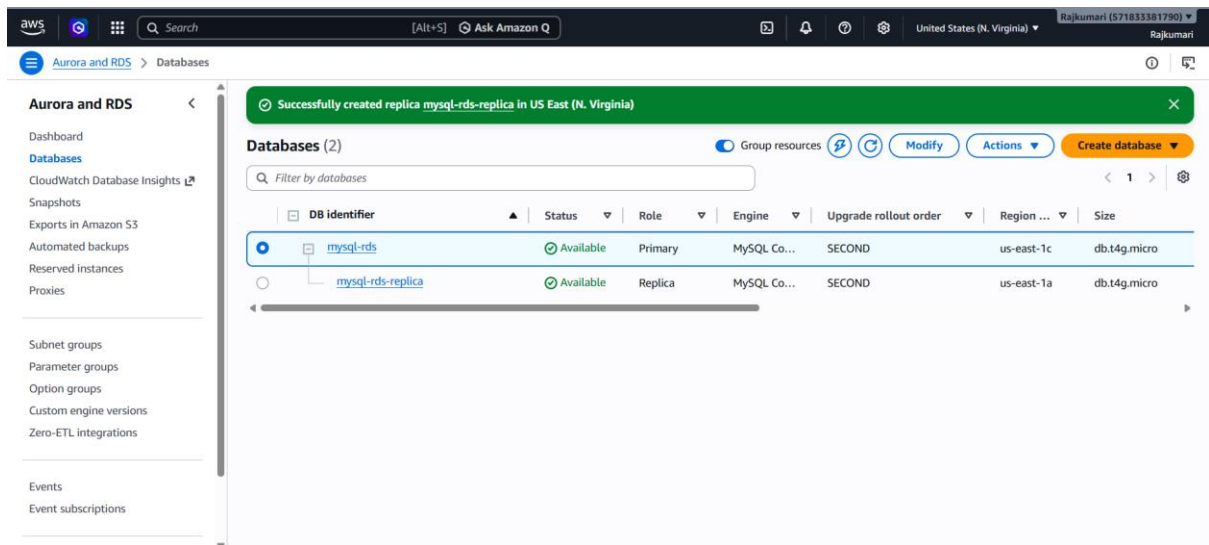
Additional storage configuration

Availability

Deployment options [Info](#)
The following deployment options are limited to those supported by the engine.

- Multi-AZ DB cluster deployment (3 instances)**
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
 - 99.95% uptime
 - Redundancy across Availability Zones
 - Increased read capacity
 - Reduced write latency
- Multi-AZ DB instance deployment (2 instances)**
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
 - 99.95% uptime
 - Redundancy across Availability Zones
- Single-AZ DB instance deployment (1 instance)**
Creates a single writer DB instance with no reader instances. This setup provides:
 - 99.5% uptime
 - No data redundancy

Connectivity



Step 7: Test the Replica

Connect to the replica:

```
mysql -h mysql-rds-replica.xxxxx.us-east-1.rds.amazonaws.com -u raji -p
```

Run:

```
SHOW DATABASES;
```

```
USE company_db;
```

```
SELECT * FROM employees;
```

The same data from the primary database should be visible.

```

root@ip-172-31-20-27:~
[root@ip-172-31-20-27 ~]# mysql -h mysql-rds-replica.cu1a004awnxp.us-east-1.rds.amazonaws.com -u raji -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.4.9 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| company_db |
| company_db_restore |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.02 sec)

mysql> USE company_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name  | department | salary |
+----+-----+-----+-----+
| 1  | Raj Kumar | DevOps      | 65000.00 |
| 2  | Priya   | HR          | 50000.00 |
| 3  | Ravi    | Development | 70000.00 |
+----+-----+-----+-----+
3 rows in set (0.02 sec)

mysql> |

```

Step 8: Verify Replication

On the **primary database**, insert a new record:

root@ip-172-31-20-27:~

```
[root@ip-172-31-20-27 ~]# mysql -h mysql-rds.cula004awnxp.us-east-1.rds.amazonaws.com -u raji -p
```

Enter password:

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 120

Server version: 8.4.9 Source distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> show databases;
```

Database
company_db
company_db_restore
information_schema
mysql
performance_schema
sys

6 rows in set (0.01 sec)

```
mysql> use company_db;
```

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> INSERT INTO employees(name,department,salary)
```

```
-> VALUES
```

```
-> ('Anjali','QA',55000);
```

Query OK, 1 row affected (0.02 sec)

```
mysql> SELECT * FROM employees;
```

id	name	department	salary
1	Raj Kumar	DevOps	65000.00
2	Priya	HR	50000.00
3	Ravi	Development	70000.00
4	Anjali	QA	55000.00

4 rows in set (0.00 sec)

```
mysql> |
```


root@ip-172-31-20-27:~

mysql> exit

Bye

[root@ip-172-31-20-27 ~]# mysql -h mysql-rds-replica.cuia004awnxp.us-east-1.rds.amazonaws.com -u raji -p

Enter password:

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 48

Server version: 8.4.9 Source distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select * from employee;

ERROR 1046 (3D000): No database selected

mysql> show databases;

Database
company_db
company_db_restore
information_schema
mysql
performance_schema
sys

6 rows in set (0.03 sec)

mysql> use company;

ERROR 1049 (42000): Unknown database 'company'

mysql> use company_db;

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

mysql> show tables;

Tables_in_company_db
employees

1 row in set (0.00 sec)

mysql> select * from employees;

id	name	department	salary
1	Raj Kumar	DevOps	65000.00
2	Priya	HR	50000.00
3	Ravi	Development	70000.00
4	Anjali	QA	55000.00

4 rows in set (0.00 sec)

mysql> |